



AAA Principles and Configuration



Foreword

- User management is one of the most basic security management requirements for any network.
- Authentication, authorization, and accounting (AAA) is a management framework that provides a security mechanism for authorizing some users to access specified resources and recording the operations of these users. AAA is widely used because of its good scalability and easy implementation of centralized management of user information. AAA can be implemented through multiple protocols. In actual applications, the Remote Authentication Dial-In User Service (RADIUS) protocol is the most commonly used to implement AAA.
- This course describes the basic concepts, implementation, basic configurations, and typical application scenarios of AAA.



Objectives

- Upon completion of this course, you will be able to:
 - Understand the fundamentals of AAA.
 - Describe the application scenarios of AAA.
 - Understand the fundamentals of RADIUS.
 - Get familiar with the basic configurations of AAA.



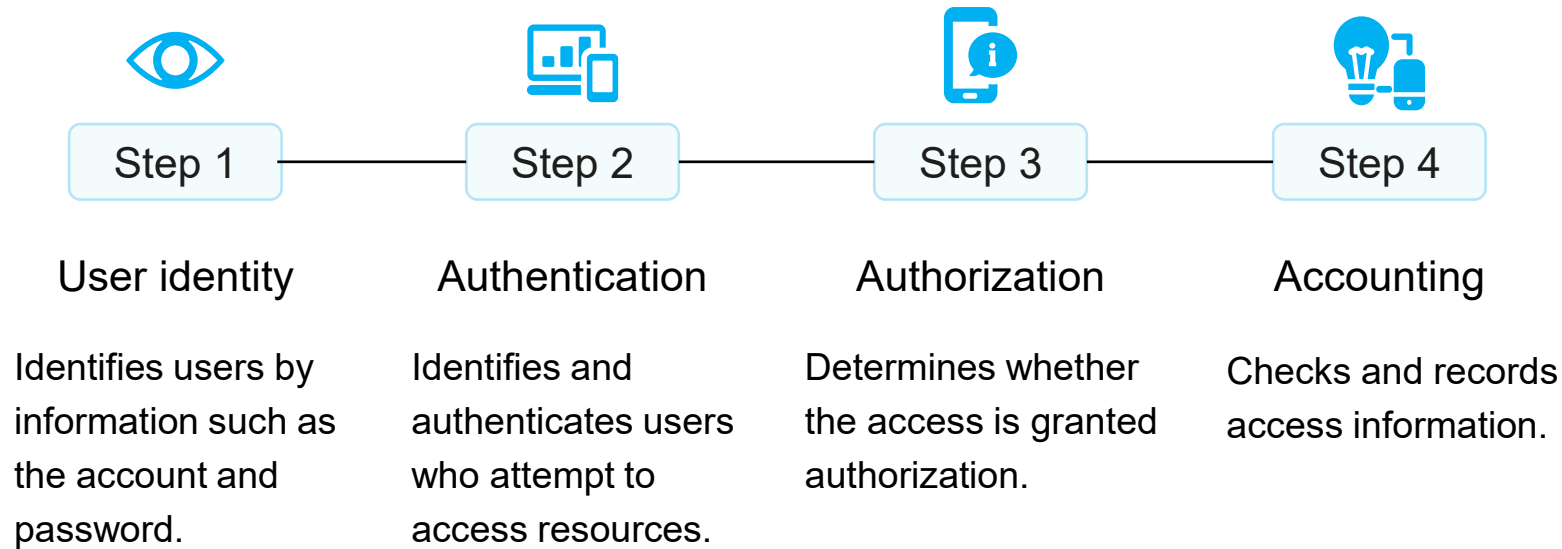
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- 1. AAA Overview**
2. AAA Configuration



Basic Concepts of AAA

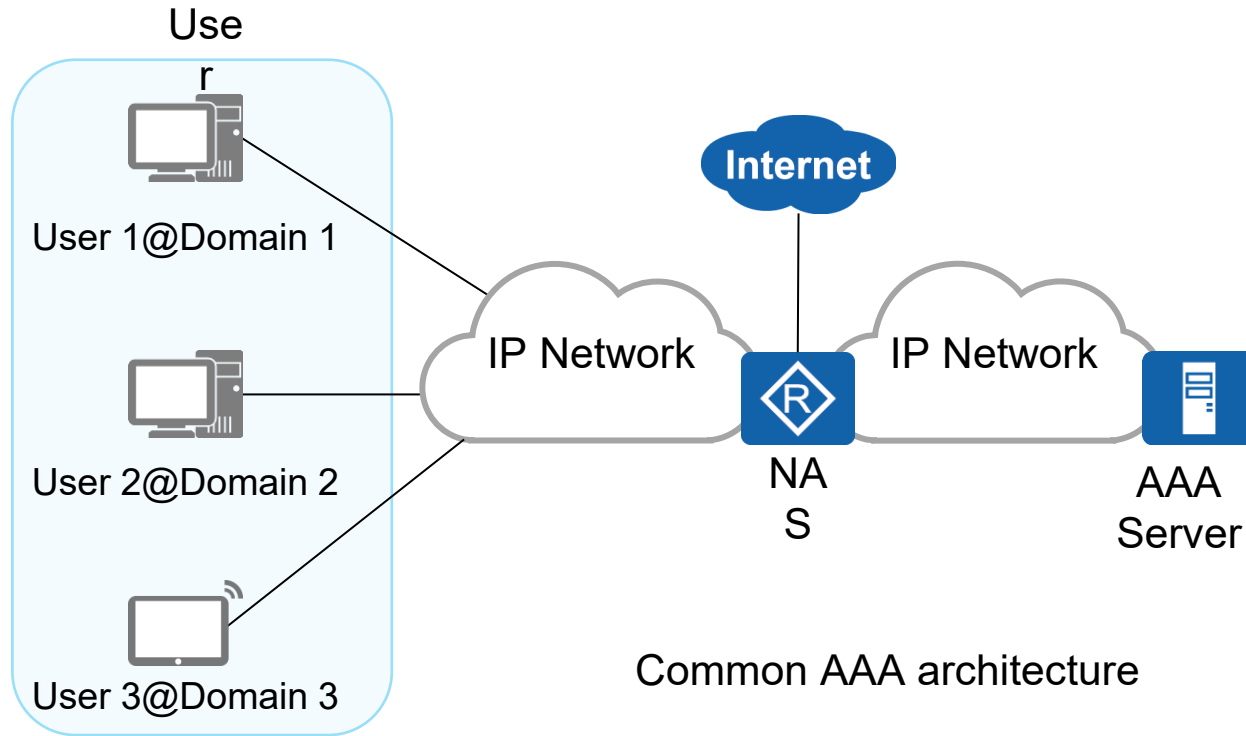
- Authentication, authorization, and accounting (AAA) provides a management mechanism for network security.





Common AAA Architecture

- A common AAA architecture includes the user, network access server (NAS), and AAA server.

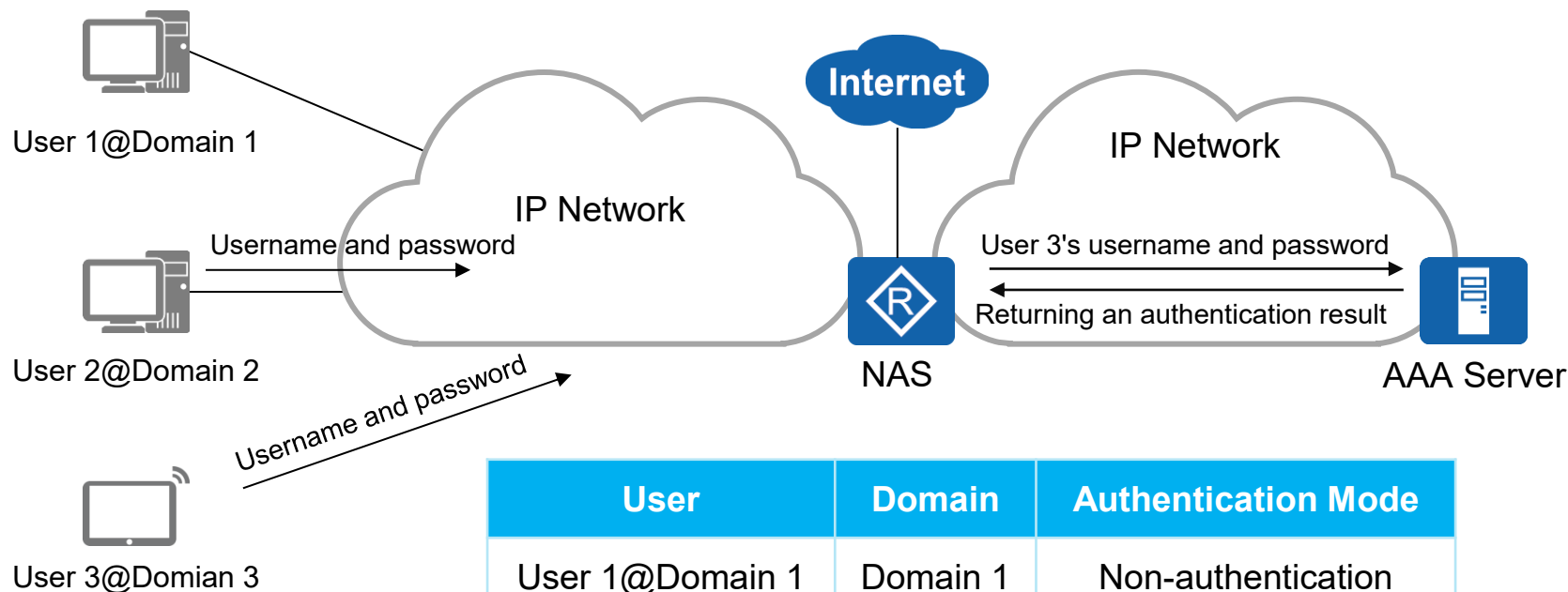


- The NAS collects and manages user access requests in a centralized manner.
- Multiple domains are created on the NAS to manage users. Different domains can be associated with different AAA schemes, which include the authentication scheme, authorization scheme, and accounting scheme.
- When receiving a user access request, the NAS determines the domain to which the user belongs based on the username and performs user management and control based on the AAA schemes configured for the domain.



Authentication

- AAA supports the following authentication modes: non-authentication, local authentication, and remote authentication.

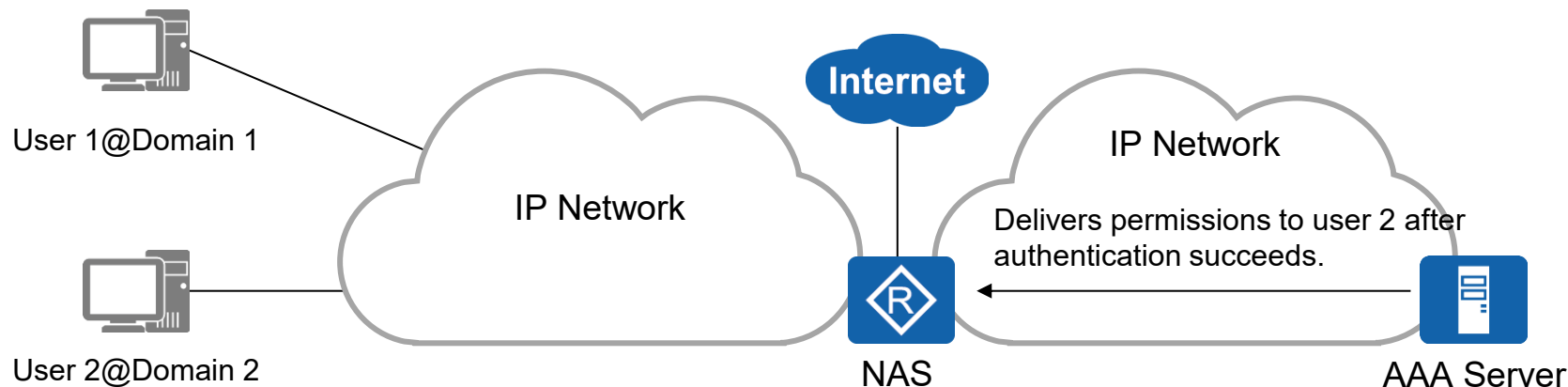


User	Domain	Authentication Mode
User 1@Domain 1	Domain 1	Non-authentication
User 2@Domain 2	Domain 2	Local authentication
User 3@Domain 3	Domain 3	Remote authentication



Authorization

- AAA supports the following authorization modes: non-authorization, local authorization, and remote authorization.
- Authorization information includes the user group, VLAN ID, and ACL number.



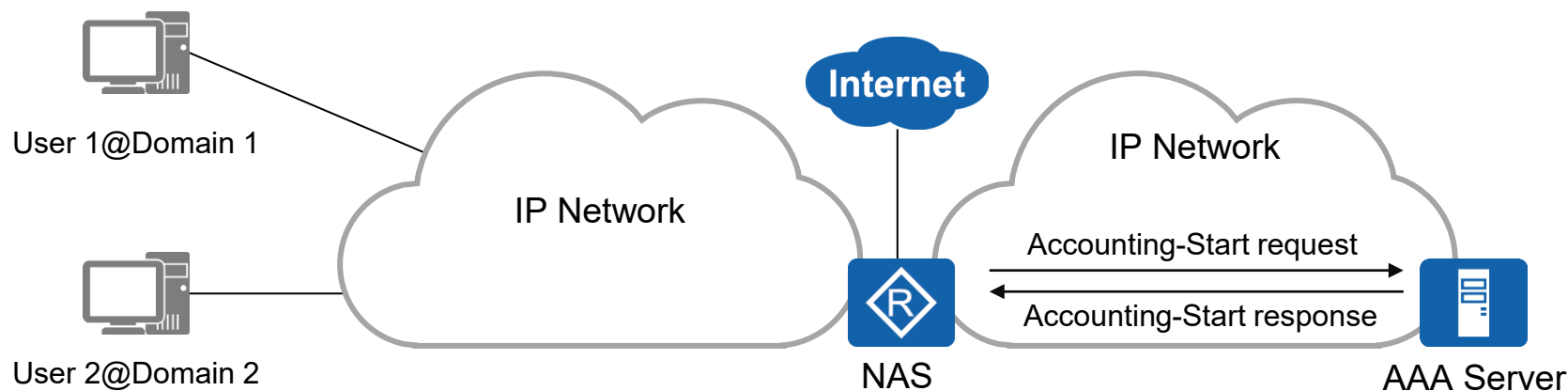
User 3@Domain 3

User	Domain	Authorization Mode	Authorization Content
User 1@Domain 1	Domain 1	Non-authorization	None
User 2@Domain 2	Domain 2	Local authorization	Internet access is allowed.
User 3@Domain 3	Domain 3	Remote authorization	Authorization is granted by a remote server.



Accounting

- The accounting function monitors the network behavior and network resource utilization of authorized users.
- AAA supports two accounting modes: non-accounting and remote accounting.

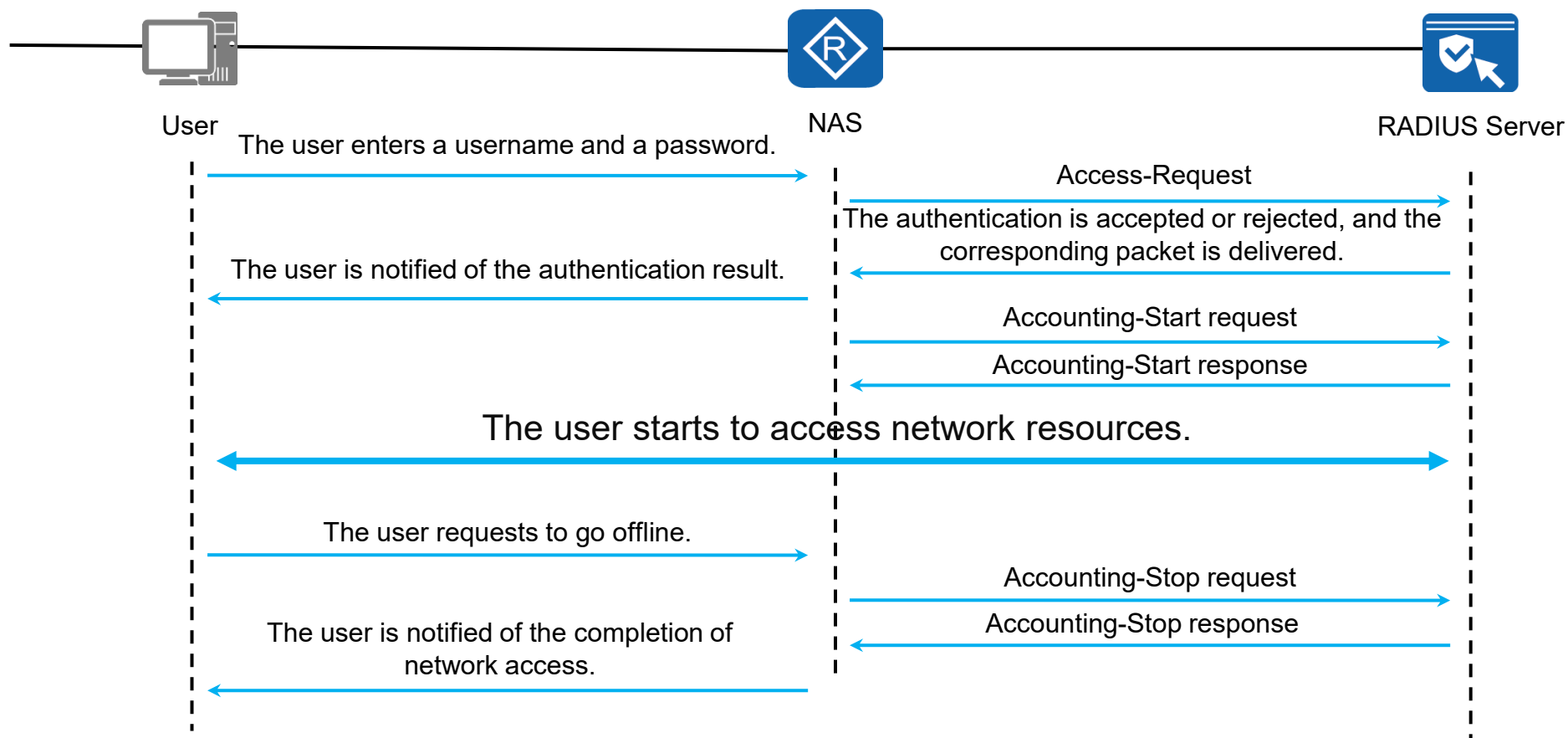


User	Domain	Accounting Mode
User 1@Domain 1	Domain 1	Non-accounting
User 2@Domain 2	Domain 2	Non-accounting
User 3@Domain 3	Domain 3	Remote accounting



AAA Implementation Protocol - RADIUS

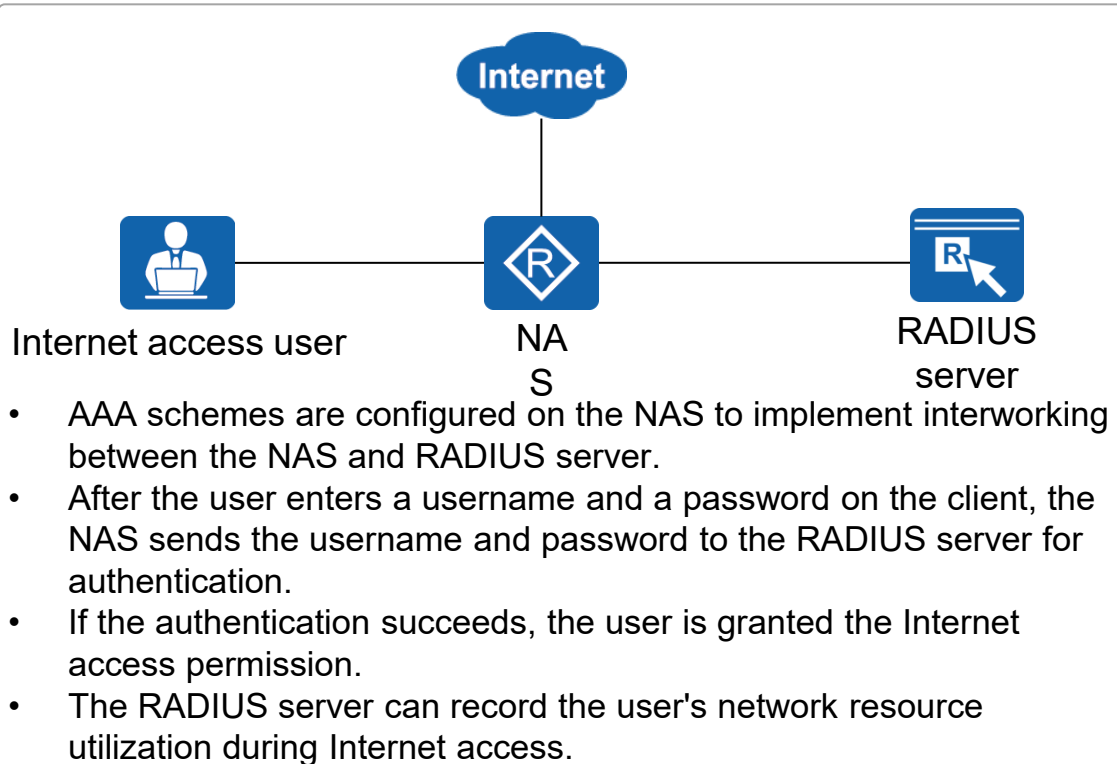
- Of the protocols that are used to implement AAA, RADIUS is the most commonly used.



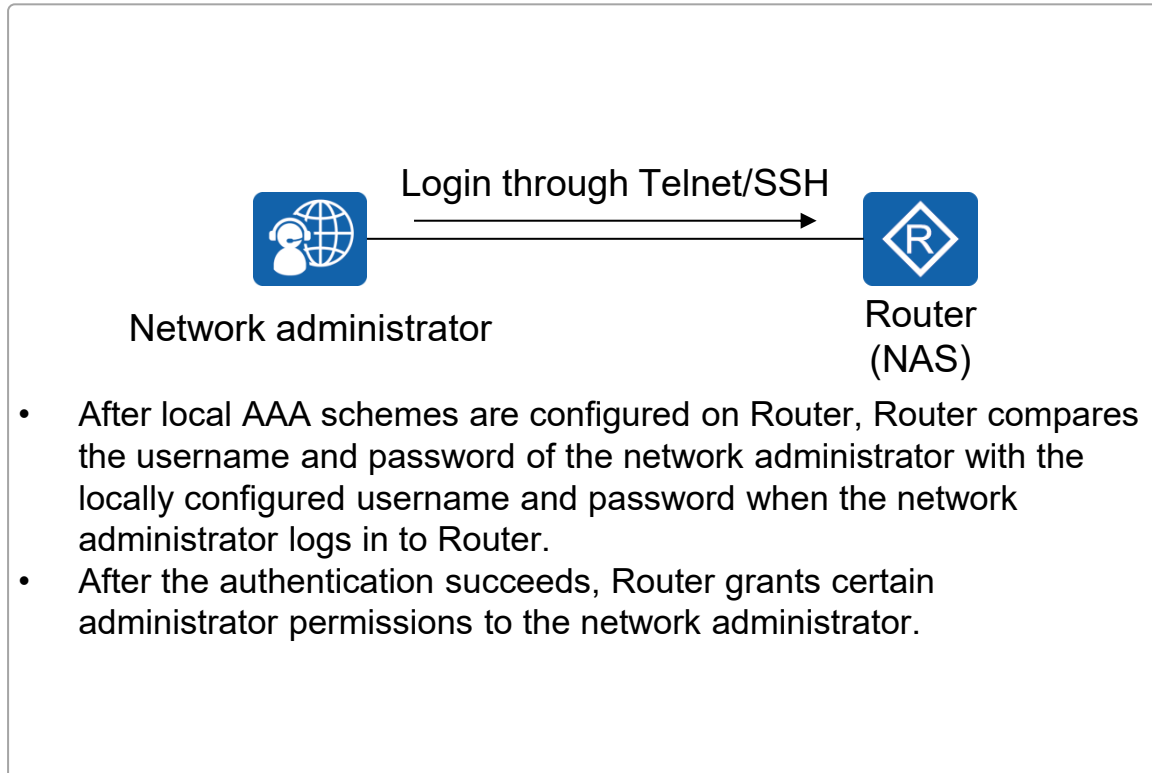


Common AAA Application Scenarios

AAA for Internet Access Users Through RADIUS



Local Authentication and Authorization for Administrative Users





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1. AAA Overview

2. AAA Configuration



AAA Configuration (1)

1. Enter the AAA view.

```
[Huawei] aaa
```

Exit the system view and enter the AAA view.

2. Create an authentication scheme.

```
[Huawei-aaa] authentication-scheme authentication-scheme-name
```

Create an authentication scheme and enter the authentication scheme view.

```
[Huawei-aaa-authentication-scheme-name] authentication-mode { hwtacacs | local | radius }
```

Set the authentication mode to local authentication. By default, the authentication mode is local authentication.



AAA Configuration (2)

3. Create a domain and bind an authentication scheme to the domain.

```
[Huawei-aaa] domain domain-name
```

Create a domain and enter the domain view.

```
[Huawei-aaa-domain-name] authentication-scheme authentication-scheme-name
```

Bind the authentication scheme to the domain.

4. Create a user.

```
[Huawei-aaa] local-user user-name password cipher password
```

Create a local user and configure a password for the local user.

- If the username contains a delimiter "@", the character before "@" is the username and the character after "@" is the domain name.
- If the value does not contain "@", the entire character string represents the username and the domain name is the default one.



AAA Configuration (3)

5. Configure a user access type.

```
[Huawei-aaa] local-user user-name service-type { { terminal | telnet | ftp | ssh | snmp | http } | ppp | none }
```

Configure the access type of the local user. By default, all access types are disabled for a local user.

6. Configure a user level.

```
[Huawei-aaa] local-user user-name privilege level level
```

Specify the permission level of the local user.



AAA Configuration Examples

- After a user password and a user level are configured on R1, host A can use the configured username and password to remotely log in to R1.



```
[R1]aaa
[R1-aaa]local-user huawei password cipher huawei123
[R1-aaa]local-user huawei service-type telnet
[R1-aaa]local-user huawei privilege level 0
[R1]user-interface vty 0 4
[R1-ui-vty0-4]authentication-mode aaa
```




Configuration Verification (1)

- In AAA, each domain is associated with an authentication scheme, an authorization scheme, and an accounting scheme. In this example, the default domain is used.

```
[R1]display domain name default_admin
```

Domain-name:	default_admin
Domain-state:	Active
Authentication-scheme-name:	default
Accounting-scheme-name:	default
Authorization-scheme-name:	-
Service-scheme-name:	-
RADIUS-server-template:	-
HWTACACS-server-template:	-
User-group:	-



Configuration Verification (2)

- After the user properly logs in and logs out, you can view the user record.

```
[R1]display aaa offline-record all
```

```
-----  
User name:          huawei  
Domain name:        default_admin  
User MAC:           00e0-fc12-3456  
User access type:    telnet  
User IP address:     10.1.1.2  
User ID:             1  
User login time:     2019/12/28 17:59:10  
User offline time:   2019/12/28 18:00:04  
User offline reason: user request to offline
```



Quiz

1. What authentication, authorization, and accounting modes are supported by AAA?
2. When a new common user is configured with local authentication but is not associated with a user-defined domain, which domain does the user belong to?



Summary

- AAA improves enterprise network security and prevents unauthorized users from logging in to enterprise networks by authenticating the identities of enterprise employees and external users, authorizing accessible resources, and monitoring Internet access behavior.
 - Authentication: determines which users can access the network.
 - Authorization: authorizes users to access specific services.
 - Accounting: records network resource utilization.
- AAA technology can be implemented either locally or through a remote server.
- Of the protocols that are used to implement AAA, RADIUS is the most commonly used.

The background of the slide features a blue-tinted image of several business professionals in a modern office. They are standing in groups, some looking at documents or devices, and their reflections are visible on the glossy floor. The overall aesthetic is professional and corporate.

Thank You
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